

Artificial Intelligence

AD203 | Lecture 1: Welcome to the Course

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Acknowledgements

The course design and content have been inspired by:

- Prof. Sudeshna Sarkar, IIT Kharagpur
- NPTEL courses offered by faculty from IIT Delhi, IIT Delhi, IISC Bengaluru
- CS188 UC barkley Course
- CS50 Harvard AI Course

Transparency

Some explanations, figures, examples, and wording may have been refined or rephrased with the assistance of Generative AI tools (e.g., ChatGPT, Claude, Gemini) under the instructor's review and verification.

“Standing on the shoulders of giants.”

Topic of Discussion

1. Getting to know the course
2. Class culture & expectations
3. Evaluation criteria
5. What is Intelligence?
6. What is Artificial Intelligence?
7. Roadmap for the semester

About the Course

AD203 — Artificial Intelligence

- **Credits:** 3-0-3 (Lecture–Tutorial–Practical)
- **Semester:** 3rd Semester, B.Tech
- **Instructor:** Dr. Alapan Kuila
- **Office Hours:** TBA
- **Email:** `alapan.cse@iiitk.ac.in`

Prerequisites

Python Programming, Data Structures, Discrete Mathematics, Basic Probability, Linear Algebra

Course Portal: [https://](https://genaicourse-dr-alapan.github.io/Artificial_intelligence/)

genaicourse-dr-alapan.github.io/Artificial_intelligence/
Syllabus, weekly slides, assignments and resources are all posted here —
bookmark it.

Why This Course?

From search algorithms to the Large Language Models you already use every day — this course builds the foundations underneath all of it.

Reason

Agents, search, logic, CSPs,
games

Learn

Probabilistic reasoning, MDPs,
RL

Scale

Deep learning, Transformers,
LLMs

Class Culture

Class Culture & Expectations

- **Be on time and be present** — attendance reflects respect for your peers and the instructor.
- **Ask questions freely** — no question is "too basic" in a fast-moving field of AI.
- **Discussion is encouraged** — concepts are best learned by debating and explaining them to each other.
- **Collaborate, but stay honest** — discuss ideas openly, but every submission must reflect your own understanding.
- **Respect the room** — minimize distractions so everyone can focus and learn.
- **Curiosity over competition** — we are building a community of learners, not just chasing marks.

- Students using laptops should sit towards the **back rows**, so that screens do not distract classmates seated behind or beside them.
- Phones should be on silent.

In this course, an LLM counts as another student —
and an open book.

- Use AI freely to build intuition, brainstorm, debug, or learn syntax
- Any use **must be disclosed** in your submission, just like a human collaborator
- You must be able to explain *every line* you submit — in code, reports, or written answers
- Undisclosed or unexplainable AI-generated submissions are treated as academic misconduct

- Attendance follows institute rules and regulations
- Collaboration and discussion are encouraged — but every submission must reflect **your own understanding**
- If you collaborate, **name your collaborators** in the submission
- Plagiarism or unauthorized copying is academic misconduct and may lead to:
 - Negative marks
 - Reporting under the institute's Academic Integrity Policy
 - Course deregistration or disciplinary action

Evaluation Criteria

How You Will Be Evaluated

Component	Weight
■ End Semester Exam	60%
■ Programming Assignments	15%
■ Mid Semester Exam	10%
■ Quiz	10%
■ Class Participation	5%

Resources You Will Use

Textbooks

- *Artificial Intelligence: A Modern Approach* — Russell & Norvig, 4th Ed. (**primary**)
- *Deep Learning* — Goodfellow et al.
- *Reinforcement Learning* — Sutton & Barto, 2nd Ed.

Software

- Python, Jupyter notebook, Google Colab
- VS Code, Git & GitHub
- NumPy, Pandas, Scikit-learn, PyTorch

What is Intelligence?

What is Intelligence?

Intelligence is the ability to:

- **Perceive** the environment
- **Learn** from experience
- **Reason** and draw inferences
- **Represent** knowledge
- **Plan** and solve problems
- **Communicate** using language
- **Adapt** to new, unseen situations
- Act **rationally** to achieve goals

Key idea

Intelligence is not one skill — it is a *collection* of abilities that work together toward achieving goals under uncertainty.

Think About This

A man looks at a photograph and says:
*“Brothers and sisters, I have none.
But this man’s father is my father’s son.”*
Whose photograph is he looking at?

Take 60 seconds. Talk to your neighbour if you like.

A Counterintuitive Fact: Moravec's Paradox

Which is harder to build a machine for?

Beating a Grandmaster at Chess

(solved decades ago)

**Recognising your grandmother's face, or
tying a shoelace**

(still genuinely hard!)

High-level reasoning needs little computation; sensorimotor skills need enormous computation.

This is why building “intelligent” machines is deceptively hard — and why AI's goals shifted over time.

What is Artificial Intelligence?

What is Artificial Intelligence?

Russell & Norvig group definitions of AI along two axes:

Thinking Humanly

Systems that think like humans
(cognitive modelling)

Thinking Rationally

Systems that think logically
(laws of thought)

Acting Humanly

Systems that act like humans
(Turing Test)

Acting Rationally

Systems that act to achieve
the best expected outcome

This course leans mainly on the **acting rationally** view — the modern, dominant paradigm in AI.

Artificial Intelligence is the design of computational agents that *perceive* their environment and take *actions* that maximise their chance of achieving their goals.

We will formalise this “agent” view — PEAS, rationality, environment types — starting next lecture.

Turing and the Turing Test

Alan Turing (1912–1954)

- British mathematician, logician, and codebreaker
- Laid the theoretical foundation of computing (the Turing Machine, 1936)
- In 1950, asked a bold question in his paper *“Computing Machinery and Intelligence”*:

Turing's Question

“Can machines think?”

The Imitation Game (Turing Test)

Turing side-stepped the philosophically messy question “can machines think?” and replaced it with a testable one: *can a machine imitate a human convincingly?*

- The judge converses (via text) with a human and a machine, without seeing either
- If the judge cannot reliably tell which is the machine, the machine **passes** the test

Is the Turing Test a Good Measure of Intelligence?

Strengths

- Operational — gives a concrete, testable criterion
- Avoids arguing about “what thinking really is”
- Historically framed the entire field’s ambition

Criticisms

- Rewards *mimicking* humans, not genuine understanding (see: Searle’s Chinese Room)
- Ignores perception, action, and embodiment
- Says nothing about *how* intelligence should be achieved

Modern AI mostly moved on from “imitate humans” to “act rationally” — but Turing’s question still frames every debate about LLMs today.

AI in the Real World

AI is Already Around You

Perception

Face unlock, speech recognition,
medical imaging

Decision-Making

Recommendation systems, route
planning, fraud detection

Language & Reasoning

Chatbots, translation, LLMs
(ChatGPT, Claude, Gemini)

Every one of these builds on ideas we will study — search, probability, learning, and reasoning.

Course Roadmap

Semester Roadmap — 6 Modules, 14 Weeks

	Module	Weeks
I	Introduction to AI — history, agents, PEAS, rationality	1–2
II	State-Space Search — BFS, DFS, UCS, A*, local search	3–5
III	Game Playing & CSPs — Minimax, α - β pruning	6–7
IV	Probabilistic Reasoning — Bayes' theorem, Bayesian networks	8–10
V	Reinforcement Learning — MDPs, Q-Learning, SARSA	11–12
VI	Modern AI — Deep Learning, Transformers, LLMs, AI Safety	13–14

By the End of This Course, You Will Be Able To...

- Design intelligent agents that perceive and act rationally
- Implement uninformed and informed search algorithms
- Solve game-playing problems and Constraint Satisfaction Problems
- Build Bayesian Networks and reason under uncertainty
- Model decision-making with MDPs and reinforcement learning
- Explain the ideas behind deep learning and Large Language Models

Questions

Before You Leave

- Visit the course portal and bookmark it:
`https://genaicourse-dr-alapan.github.io/Artificial\_intelligence/`
- Set up Python, Jupyter/Colab, and a GitHub account before next lecture
- Read: Russell & Norvig, Chapter 1
- Office Hours: TBA

Questions?